

JEOL01 DATA FORMAT SPECIFICATION

This manual applies to the following
JEOL products:

JBX-6300 series	JBX-5000LS
JBX-5000LS/E	JBX-6000FS
JBX-6000FS/E	JBX-9300FS
JBX-5500 series	

For the proper use of the instrument, be sure to read this instruction manual. Even after you read it, please keep the manual on hand so that you can consult it whenever necessary.

JEOL01 DATA FORMAT SPECIFICATION

**Please be sure to read this instruction manual carefully,
and fully understand its contents prior to the operation
or maintenance for the proper use of the instrument.**

NOTICE

- The following actions must be avoided without prior written permission from JEOL Ltd. or its subsidiary company responsible for the subject (hereinafter referred to as "JEOL"): modifying the instrument; attaching products other than those supplied by JEOL; repairing the instrument, components and parts that have failed, such as replacing pipes in the cooling water system, without consulting your JEOL service office; and adjusting the specified parts that only field service technicians employed or authorized by JEOL are allowed to adjust, such as bolts or regulators which need to be tightened with appropriate torque. Doing any of the above might result in instrument failure and/or a serious accident. If any such modification, attachment, replacement or adjustment is made, all the stipulated warranties and preventative maintenances and/or services contracted by JEOL or its affiliated company or authorized representative will be void.
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Note: For servicing and inquiries, please contact your JEOL service office.

NOTATIONAL CONVENTIONS AND GLOSSARY

■ Examples for general notations

- **CAUTION** – : Important precautions for use, which, if not followed, may result in damage to or problems with the device itself.
-  : Additional points to remember regarding the operation.
-  : A reference to another section, chapter or manual.
- 1, 2, 3** : Numbers indicate a series of operations that achieve a task.
-  : A diamond indicates a single operation that achieves a task.
- File**: The names of menus, commands, or parameters displayed on the screen are denoted with **bold** letters.
- File–Exit** : Selecting a menu item from a pulldown menu is denoted by linking the menu and the item with a dash (–).
For example, **File–Exit** means selecting **Exit** from the **File** menu.
-  : Keys on the keyboard are denoted by enclosing their names in a box.

■ Examples for mouse terminology

- Mouse pointer: A mark, displayed on the screen, which moves following the movement of the mouse. It is used to specify a menu item, command, parameter value, and other items. Its shape changes according to the situation.
- Click: To press and release the left mouse button.
- Right-click: To press and release the right mouse button.
- Double-click: To press and release the left mouse button twice quickly.
- Drag: To hold down the left mouse button while moving the mouse.

CONTENTS

SAFETY PRECAUTIONS

1	GENERAL	1
2	DATA FILE SPECIFICATIONS	2
2.1	Configuration of Magnetic Disk Data	2
2.2	Configuration of Magnetic Tape Data	2
2.3	Configuration of Data Records.....	3
3	PATTERNS AND COORDINATES	4
4	COMMANDS AND DATA.....	5
4.1	ID Command for Pattern Data (ID/).....	5
4.2	Rectangular Command (RT).....	5
4.3	Polygon Command (TK)	6
4.4	Straight Line Command (LN)	8
4.5	Ring Command (RG)	8
4.6	Pattern Repetition Commands (RL & RE)	9
4.7	Data End Command (EF)	11
4.8	Comment Line Command (*)	11
4.9	Multivolume Continuation Command (CT).....	12

SAFETY PRECAUTIONS

Although this instrument is protected with safety device which prevents the occurrence of accident that could result in an injury, harm, and damage to the users or instrument itself, the safety feature may not work properly if you use the instrument for the purpose of use not intended or in an improper usage. For the proper use of the instrument, please be sure to read all of the instructions, descriptions, notices, and precautions contained in this manual carefully to understand them fully prior to the operation or maintenance. This section, "Safety Precautions," contains important information related to safety for using of the instrument.

The safety indications and their meanings are as follows:

	DANGER: An imminently hazardous situation which, if not avoided, will result in death or serious injury.
	WARNING: A potentially hazardous situation which, if not avoided, could result in death or serious injury.
	CAUTION: A potentially hazardous situation which, if not avoided, may result in minor or moderate injury, or a situation that could result in serious damage to facilities or acquired data.

Labels bearing the following symbols are attached to dangerous locations on the instrument. Do not touch any of these locations with your hands or anything else.



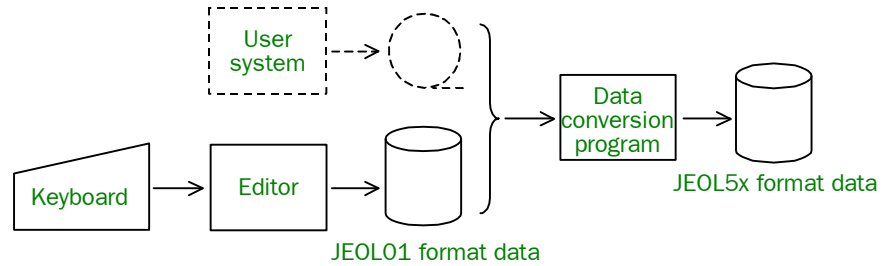
Examples of symbols

-  • Use the instrument properly within the scope of the purpose and usage described in its brochures and manuals.
-  • Never open/remove protective parts (exterior panels) and parts that can't be opened/removed without use of tool (including key), or disconnect/connect the cables/connectors that are not described in this manual.
-  • Never attempt to do any works of disassembling/assembling the instrument other than those described in this manual.
-  • Never make modifications that include installing substitute parts and disabling safety devices or other safety features.
-  • Never disconnect the grounding wire or move it from the prescribed position. Failure to follow this instruction could result in electric shock.
-  • To avoid falling, do not climb onto the operation table and console during daily operation or during maintenance or inspection.
-  • When you dispose of the instrument or liquid or other waste, follow all applicable laws and regulations, and dispose of it in a proper manner without polluting the environment.
-  • Be sure to read the "Safety Precautions" section of the manuals for the accessories attached to or built into the instrument.
-  • If anything is unclear, please contact your JEOL service office.

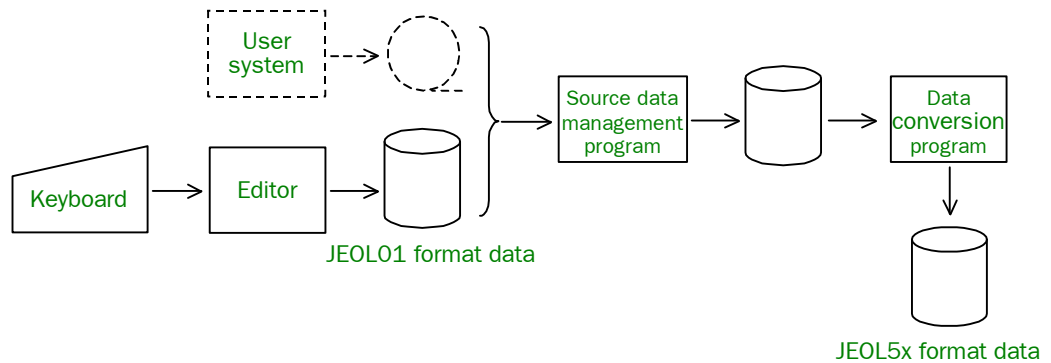
1 GENERAL

This manual describes the specifications for data files that are applied to write patterns on substrates using a JEOL charged particle beam lithography system. The data files are stored on disk or on magnetic tape, and the format used is the JEOL01 data format. Please note that there are two types of data conversion processes for JEOL01 format data. The process used depends on the computer system, as shown below.

- When data conversion is performed with a PDP computer system



- When data conversion is performed with a VAX or a UNIX computer system



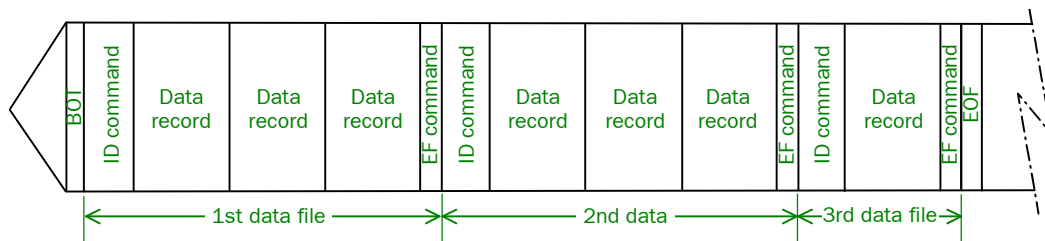
2 DATA FILE SPECIFICATIONS

2.1 Configuration of Magnetic Disk Data

- Data is written in each file on the disk.
- Any disk filename can be used, as long as the computer system that is controlling the disk accepts the name. The filename extension is also optional, but in general “.J01” is used.
- A file is composed of one or more formatted sequential data files.
- A data file begins with the ID command, contains one or more data records, and ends with the EF command.
- Each data file is identified by the pattern data ID name designated in its ID command.
- A data record consists of variable-length records of up to 511 bytes.
- The record attribute is a carriage return/carriage control.
- The contents of a file are described by using 7-bit ASCII codes.

2.2 Configuration of Magnetic Tape Data

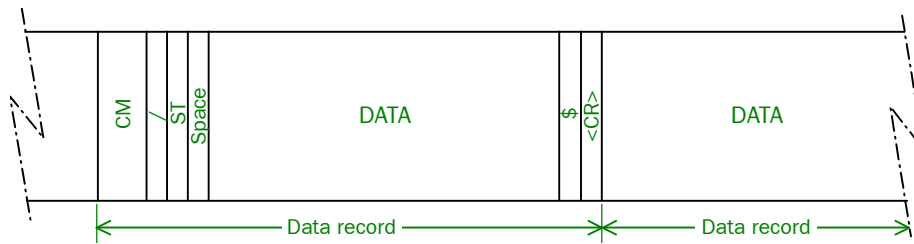
- Magnetic tape data begins with a BOT (Beginning-Of-Tape) mark, contains one or more data files, and ends with an EOF (End-Of-File) mark or the tape mark.
- A data file begins with the ID command, contains one or more data records, and ends with the EF command, as shown below.



- Each data file is identified by the pattern data ID name designated in its ID command.
- A data record consists of 20 – 512 characters. Space codes are used as filler when the length is shorter than 20 characters.
- The contents of a file are described by using 7-bit ASCII codes.

2.3 Configuration of Data Records

A data record is composed of the items shown below.



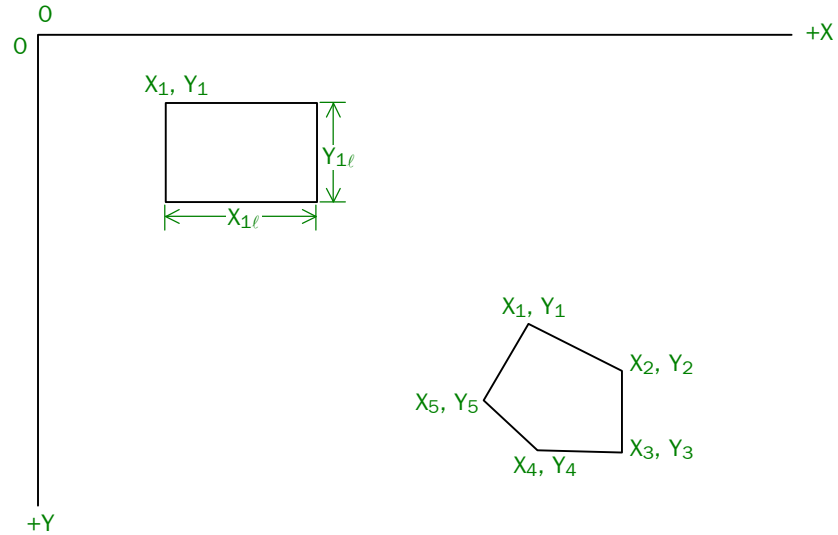
“CM”/”ST”△“DATA”\$<CR>

- “CM”: Two characters; command (RT, TK, etc.)
- /: Slash; delimits command and ST.
- “ST”: One or two characters; scan speed modulation code (0 – 15, 0 – 63, or 0 – 1023), depends on instrument and command. The range 0 – 1023 is used only in the JBX-9300FS.
- △: Space
- “DATA”: Variable length characters; parameters (coordinates, pattern size, etc.)
- :: Semicolon; delimits data, and it is used in the rectangular and trapezoid data.
- \$: Dollar mark; this signifies that the data continues on the succeeding line. If the data is completed in a single line, this mark is not used. (In the format shown above, the data is continuous.)
- <CR>: Carriage return

Example: RT/5△200, 100, 50, 50<CR>

3 PATTERNS AND COORDINATES

Patterns are expressed using the coordinate system in the fourth quadrant. The origin of the coordinates is at the upper left corner as shown the figure below. (unit: $0.1\ \mu\text{m}$)



4 COMMANDS AND DATA

4.1 ID Command for Pattern Data (ID/)

This command specifies the identifier of a pattern data file. Since a data file is composed of an ID command, data, and an EF command, this command is always required at the head of a data file.

ID/ABCDEFGF

ABCDEFGF: Identifier of a pattern data file. The name whose character string is 24 characters or shorter can be used.

4.2 Rectangular Command (RT)

This command serves to plot rectangles that have sides parallel to the X and Y axis.

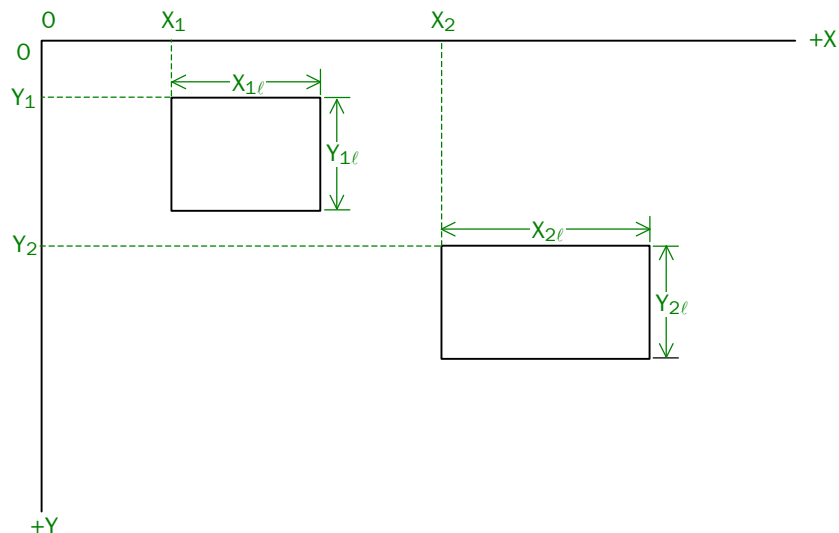
RT/ST△X₁, Y₁, X_{1ℓ}, Y_{1ℓ}; X₂, Y₂, X_{2ℓ}, Y_{2ℓ}; X₃...

Rectangular 1

Rectangular 2

Delimiter

ST: A rank number shot time modulation (0 – 15, 0 – 63, or 0 – 1023); however, 0 – 1023 is only for the JBX-9300FS.
 ;; Delimiter for each rectangular



~~✗~~ The delimiter mark “;” cannot be used at the end of the line.

(Correct)

RT/0△10, 20, 30, 40 ; 50, 60, 70, 80
 RT/0△90, 100, 110, 120

(Incorrect)

RT/0△10, 20, 30, 40 ; 50, 60, 70, 80 ;
 90, 100, 110, 120

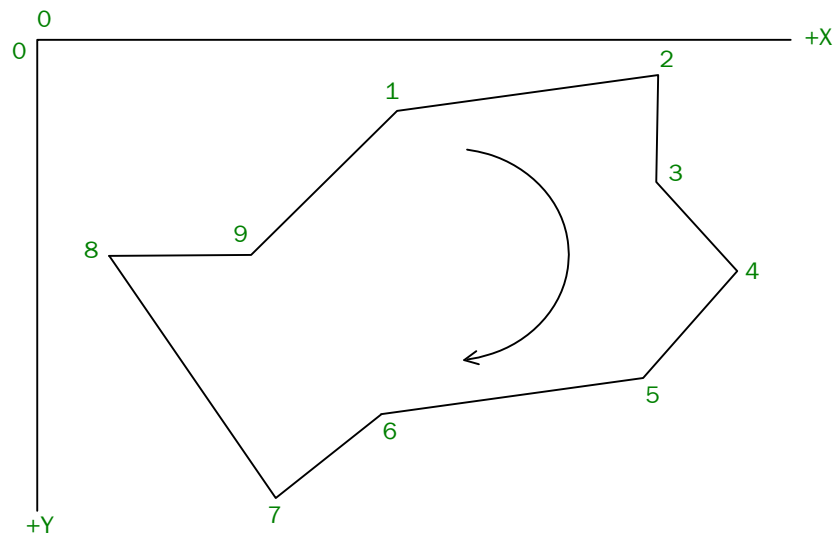
~~✗~~ The continuation mark “\$” cannot be used in this data.

4.3 Polygon Command (TK)

This command serves to plot polygons with data expressed as a group of corner coordinates arranged clockwise.

TK/ST△X₁, Y₁ – X₂, Y₂ – X₃, Y₃ – X₄, Y₄\$
 X₅, Y₅ – X₆, Y₆ – X₇, Y₇\$
 X₈, Y₈ – X₉, Y₉

ST : A rank number shot time modulation (0 – 15, 0 – 63, or 0 – 1023);
 however, 0 – 1023 is only for the JBX-9300FS.
 –: Delimiter between two corner coordinates
 \$: Denotes continuation of data



-  The corner coordinates must be ordered clockwise.
-  Up to 120 corner coordinates can be designated.
-  When data cannot be written in a line, you can continue the rest of the data in the succeeding lines by using “\$” marks. However, X_i and Y_i cannot be divided into two lines.

(Correct)

TK/0△X₁, Y₁ – X₂, Y₂\$
 X₃, Y₃ – X₄, Y₄

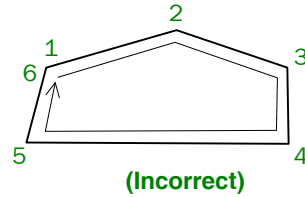
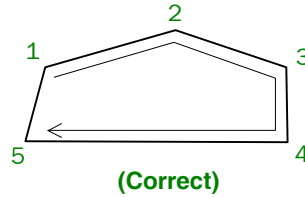
(Incorrect)

TK/0△X₁, Y₁ – X₂, Y₂ – X₃\$
 , Y₃ – X₄, Y₄

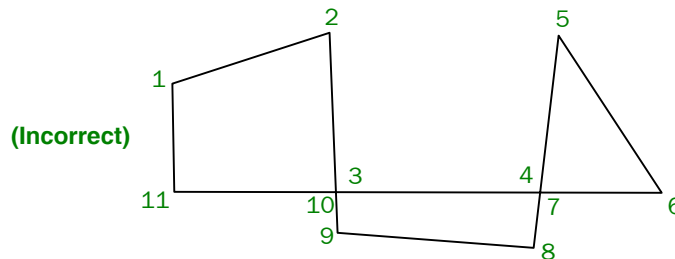
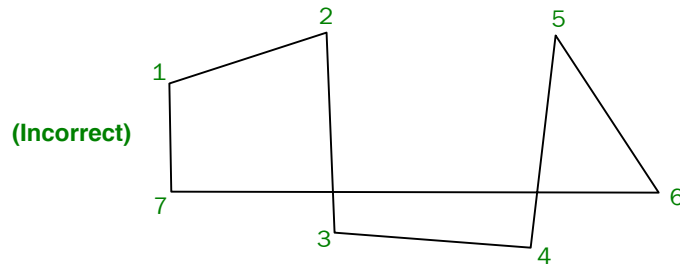
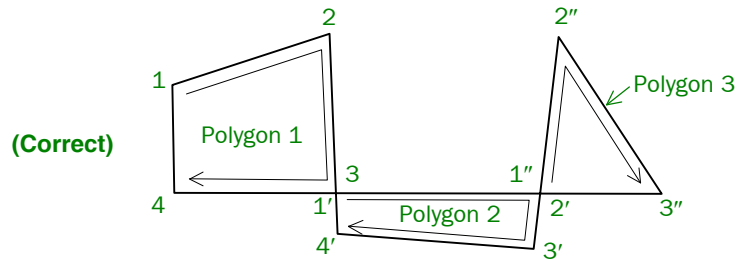
-  The last corner coordinates to be designated are those immediately before the start (first) corner coordinates.

(Correct) TK/0△X₁, Y₁ – –X₄, Y₄ – X₅, Y₅

(Incorrect) TK/0△X₁, Y₁ – –X₄, Y₄ – X₅, Y₅ – X₆, Y₆



-  Line segments that intersect each other are not allowed.



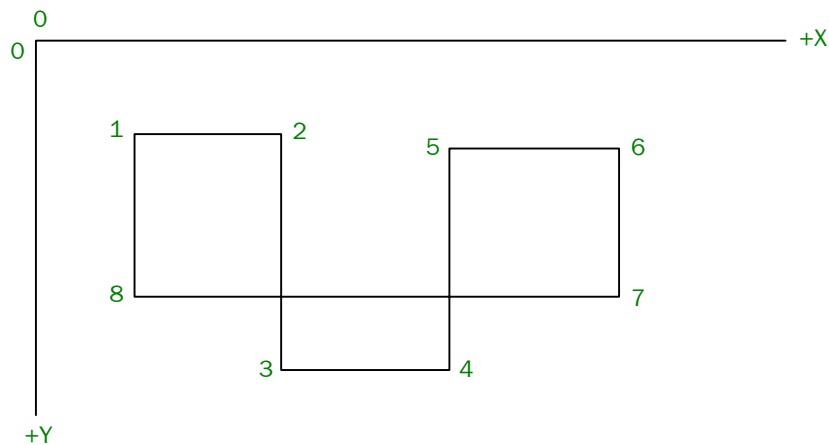
-  A semicolon (;) mark can be used as a delimiter in a polygon data expression as in rectangular data. However, the mark cannot be placed at the head or end of a line.

TK/0△X₁, Y₁ – X₂, Y₂ – X₃, Y₃ ; X₁₀, Y₁₀ – X₁₁, Y₁₁\$
X₁₂, Y₁₂ – X₁₃, Y₁₃ ; X₂₀, Y₂₀ – X₂₁, Y₂₁ – X₂₂, Y₂₂

4.4 Straight Line Command (LN)

This command serves to plot straight lines that have a width of one point and are parallel to the X and Y axis. The expression of data is similar to that of the polygon data, however, straight lines are plotted at each corner. This means that the straight line data uses expressions where, for example, line segments intersect and the start and end coordinates coincide, although these are not allowed in polygon data.

LN/ST△X₁, Y₁ – X₂, Y₂ – X₃, Y₃ – X₄, Y₄\$
X₅, Y₅ – X₆, Y₆ – X₇, Y₇ – X₈, Y₈ – X₁, Y₁



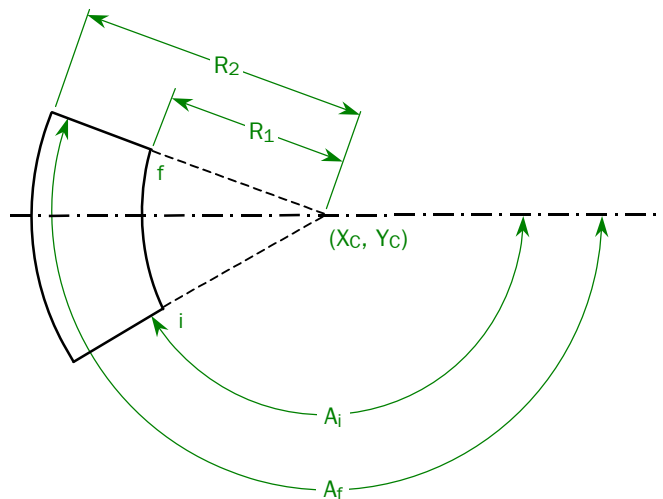
-  Up to 120 right angle corners can be designated.
-  A “\$” mark can be used in this data. The application method is the same as that in polygon data.
-  This command cannot be used with the variable shaped electron beam lithography system (i.e., JBX-7000MV).
-  The range of ST for a single line is 0 – 15 or 0 – 1023; however, 0 – 1023 is only for the JBX-9300FS.

4.5 Ring Command (RG)

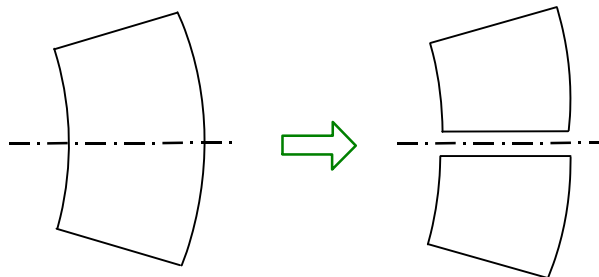
This command serves to plot rings and discs. The ring data is expressed by using the central coordinates, inside radius, outside radius, initial angle, and final angle of the ring.

RG/ST△X_C, Y_C, R₁, R₂, A_i, A_f

- | | |
|-----------------------------------|--|
| ST: | A rank number shot time modulation (0 – 15, 0 – 63, or 0 – 1023); however, 0 – 1023 is only for the JBX-9300FS. |
| X _C , Y _C : | Center coordinates |
| R ₁ : | Inside radius |
| R ₂ : | Outside radius R ₂ ≥ R ₁ ≥ 0 |
| A _i , A _f : | Angles (unit: one degree)
360 > A _i ≥ 0, 360 ≥ A _f > 0, 45 ≥ (A _f – A _i) > 0 |



 Divide it into two patterns when the pattern is as shown below.



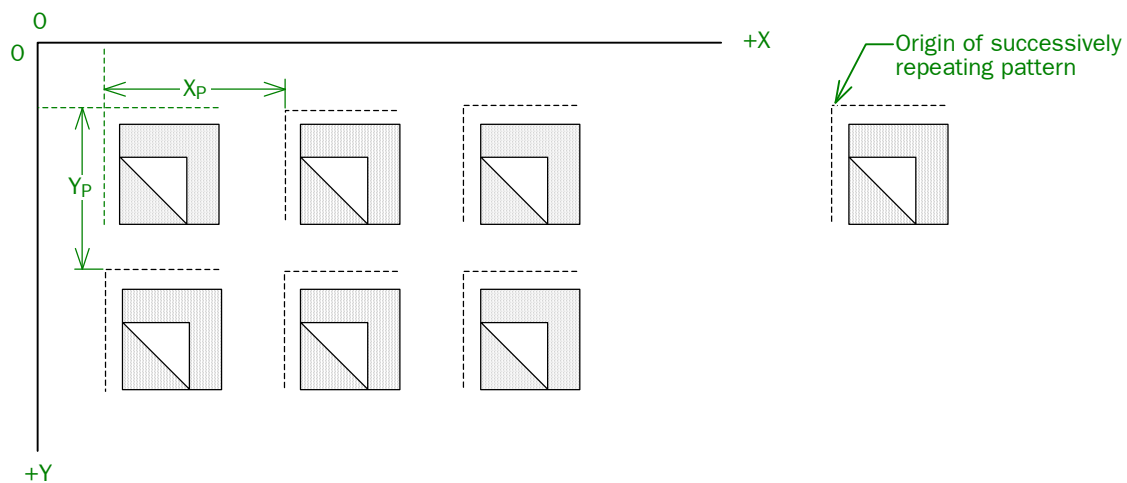
-  This command cannot be used with the variable shaped electron beam lithography system (i.e., JBX-7000MV).
-  The data describes a disc when R1 is 0.
-  The patterns written by ring data are composed of groups of rectangular patterns and/or trapezoidal patterns.

4.6 Pattern Repetition Commands (RL & RE)

These commands serve to repeat pattern writing of the pattern data that is sandwiched between the RL and RE command lines.

```

RL△X, Y, XP, YP, XN, YN
Pattern data (RT, TK or similar command)
.
.
.
.
RE
  
```



X,Y: Origin coordinates of repeating pattern

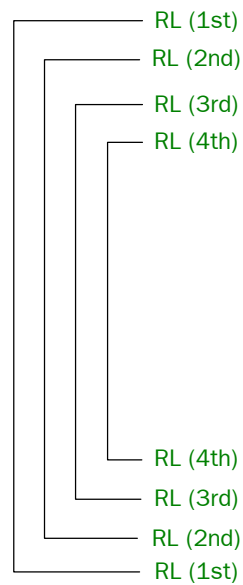
XP: Pitch of repetition in the X direction

YP: Pitch of repetition in the Y direction

XN: Number of repetition in the X direction $1 \leq XN \leq 30000$

YN: Number of repetition in the Y direction $1 \leq YN \leq 30000$

 Nesting for repetition commands is allowed up to 4 times.



4.7 Data End Command (EF)

This command indicates the end of a data file and is always used with an ID command. That is, whenever a data file is started with an ID command, it is terminated with an EF command. However, in the case of data on a magnetic tape (MT), the tape mark can also act as a terminator.

The following is an example of a data format that includes multiple ID names and data.

```
ID/NAME1
.
.
.
.
EF
ID/NAME2
.
.
.
.
EF
ID/NAME3
.
.
.
.
EF
```

4.8 Comment Line Command (*)

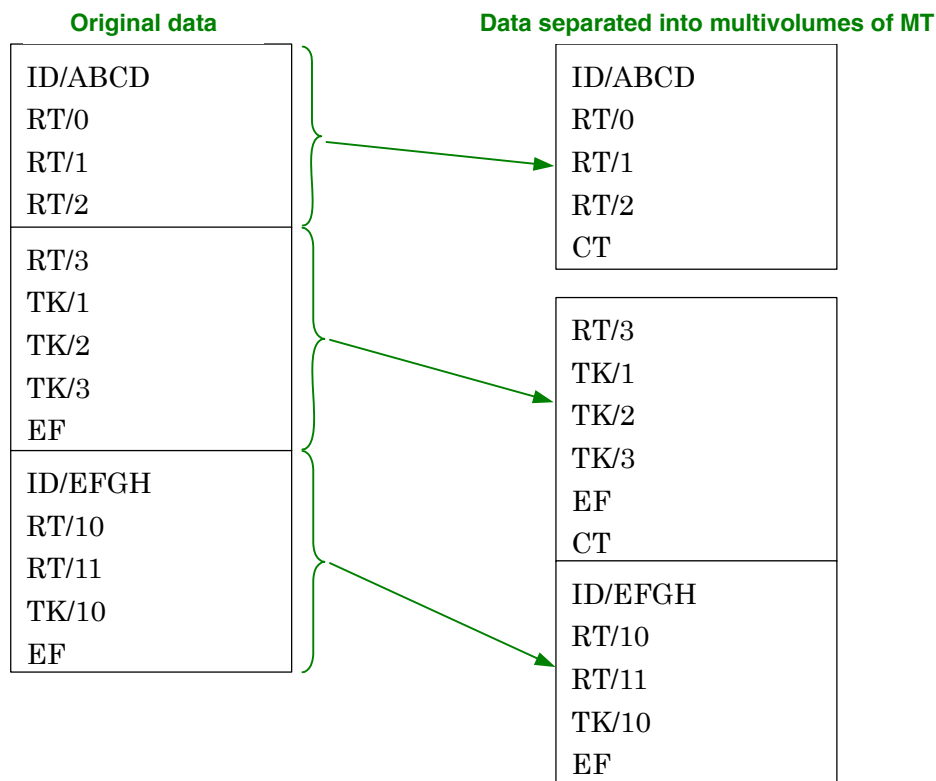
When you wish to add comments to data, write an asterisk (*) mark followed by the comments, starting from the first column of a line.

```
*△△SAMPLE ← Comment line
ID/ABCD
RT/0△10, 20, 30, 40
*△△△△REPEAT ← Comment line
RL△50, 200, 50, 50, 2, 2
RT/0△40, 50, 60, 70
RE
*△△△△REPEAT△END ← Comment line
EF
```

4.9 Multivolume Continuation Command (CT)

This command serves to divide data into multivolumes of MT (magnetic tape). When one tape medium cannot store all the pattern data, CT command lines are added to the pattern data. The CT command must be written starting at the 1st column in the line similar to a RT or TK command.

Example:



- ~~✗~~ This command is exclusively used in files on magnetic tape media. It cannot be used in files on magnetic disks.
- ~~✗~~ Data that is written between ID and EF command lines can be divided into two or more volumes of MT by CT commands.
- ~~✗~~ TK command data that is continued with a \$ mark cannot be divided using a CT command line.

(Correct)

```
TK/1△X1, Y1 - ..... $
Xn, Yn - ..... $
Xm, Ym - .....$
CT
TK/2
```

(Incorrect)

```
TK/1△X1, Y1 - ..... $
Xn, Yn - ..... $
CT
Xm, Ym - .....
TK/2
```

- ~~✗~~ EOF and other commands can follow a CT command line. End check is made at the CT command line when the MT is read.
- ~~✗~~ Data must be divided using a CT command before the EOT (End of tape) mark is detected.